

Question	Marking Guidance	Mark	Comments
10.1	1. A <u>metabolite</u> in condensation/hydrolysis/ photosynthesis/respiration; 2. A solvent so (metabolic) reactions can occur OR A solvent so allowing transport of substances; 3. High heat capacity so buffers changes in temperature; 4. Large latent heat of vaporisation so provides a cooling effect (through evaporation); 5. Cohesion (between water molecules) so supports columns of water (in plants); 6. Cohesion (between water molecules) so produces surface tension supporting (small) organisms;	5 max	3. For 'buffer' accept 'resist'. 5. For 'columns of water' accept 'transpiration stream'. Do not credit 'transpiration' alone but accept description of 'stream'. 5. For 'columns of water' accept 'cohesion-tension (theory)'. 5 and 6. For cohesion accept hydrogen bonding Ignore reference to pH. Allow other suitable properties but must have a valid explanation. For example - ice floating so maintaining aquatic habitat beneath - water transparent so allowing light penetration for photosynthesis

Question	Marking Guidance	Mark	Comments
10.2	<u>Lipid</u> 1. Add ethanol/alcohol then add water and shake/mix OR Add ethanol/alcohol and shake/mix then pour into/add water; 2. White/milky <u>emulsion</u> OR <u>emulsion</u> test turns white/milky; <u>Non-reducing sugar</u> 3. Do Benedict's test and stays blue/negative; 4. <u>Boil</u> with acid then neutralise with alkali; 5. Heat with Benedict's and becomes red/orange (precipitate); <u>Amylase</u> 6. Add biuret (reagent) and becomes purple/violet/mauve/lilac; 7. Add starch, (leave for a time), test for reducing sugar/absence of starch;	5 max	4 max if marks gained from only 2 substance tests. 1. Reject heating emulsion test. 1. Accept 'Add Sudan III and mix'. 2. Ignore cloudy. 2. Reject precipitate. 2. Accept (for Sudan III) top (layer) red. 3. Ignore details of method for Benedict's test for this mp. 4. Accept named examples of acids/alkalis. 5. Do not credit mp5 if no attempt at mp4. 5. For 'heat' ignore 'warm'/'heat gently'/'put in a water bath' but accept stated temperatures $\geq 60^{\circ}\text{C}$. 5. Heat must be stated again, do not accept using residual heat from mp4. 5. Accept 'do the Benedict's test' if full correct method given elsewhere. 5. Accept 'sodium carbonate, sodium citrate and copper sulfate solution' for Benedict's but must have all three if term 'Benedict's' not used. 6. Accept 'sodium or potassium hydroxide and copper sulfate solution' for 'biuret'. 6. Reject heating biuret test.

Question	Marking Guidance	Mark	Comments
10.3	1. A condensation reaction joins monomers together and forms a (chemical) bond and releases water; 2. A hydrolysis reaction breaks a (chemical) bond between monomers and uses water; 3. A suitable example of polymers and the monomers from which they are made; 4. A second suitable example of polymers and the monomers from which they are made; 5. Reference to a correct bond within a named polymer;	5	Ignore reference to dimers. 3. and 4. Polymers must contain many monomers. 3. and 4: suitable examples include • amino acid and polypeptide, protein, enzyme, antibody or specific example • nucleotide and polynucleotide, DNA or RNA • <u>Alpha</u> glucose and starch/glycogen • <u>Beta</u> glucose and cellulose. If neither specific carbohydrate example is given, allow monosaccharide/glucose and polysaccharide. 3. and 4. Reject (once) reference to triglycerides. 5. Reject reference to ester bond.